

HACKOUT'26 PROJECT

Team Name: FALCON	Chosen Theme: Renewable Energy Intelligence
Problem Statement: Predictive Maintenance for Solar & Wind Assets	Team Members: Harshul Jindal, Amish Dungarwal , Ansul Bhushan

1. Problem Statement

Solar panels and wind turbines operate continuously in harsh outdoor environments, facing dust accumulation, extreme thermal variations, structural vibrations, and unpredictable weather conditions. Over time, minor operational issues—such as panel soiling, elevated gearbox temperatures, subtle mechanical vibrations, or bearing wear—gradually escalate into severe equipment failures.

The primary challenge in the industry is that these faults are typically detected **reactively**—only after generation performance significantly degrades or an asset shuts down entirely.

For maintenance teams, relying on manual field inspections is **time-consuming, expensive, and difficult to execute at scale**. Technicians often travel vast distances across renewable sites to inspect operational assets that require no servicing, while undetected micro-faults elsewhere quietly worsen.

When an unexpected component failure occurs, the operational chain reaction is immediate: [Unplanned Equipment Shutdown] —> [Reduced Energy Generation] —> [Direct Financial & Revenue Loss]

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2. Proposed Solution:

We propose an AI-powered predictive maintenance platform designed for solar farms and wind farms. We collect sensor data such as temperature, vibration, current, voltage, and power output. Our AI detects abnormal behavior and predicts which assets are likely to fail. Instead of simply generating alerts, our system:

- Explains why an asset is at risk (e.g., thermal stress from dust or bearing wear from vibration).
- Assigns it a real-time health score (NORMAL, WARNING, CRITICAL).
- Prioritizes maintenance based on both failure probability and financial impact.
- Estimates energy and revenue lost due to potential downtime.

This allows renewable-energy operators to move seamlessly from reactive maintenance to proactive, data-driven maintenance.

3. Approach & Methodology:

1. Data Simulation (Lightweight & Hardware-Free)

- **How it works:** Rather than risking hardware failures on stage, we use a Python script (**pandas** & **numpy**) to simulate live IoT sensor streams for solar panels and wind turbines.
- **Monitored Parameters:**
 - **Solar:** Panel soiling (%), surface temperature (°C), current (A), voltage (V), and total power output (kW).
 - **Wind:** Turbine vibration (Hz), gearbox temperature (°C), oil pressure (bar), turbine RPM, and external wind speed (m/s).
- **Realistic Data Behavior:** The simulated numbers follow straightforward physical relationships—for instance, as panel soiling increases, current output naturally drops; as gearbox temperature rises, oil pressure drops.

2. Anomaly Detection & False-Alarm Filter

- **Threshold Checks:** Incoming sensor values are continuously compared against normal operating ranges.

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- **Smart Weather Check (Prevents False Alarms):** To avoid sending field crews on unnecessary trips due to weather noise (like a sudden wind gust causing temporary turbine vibration), the Python code cross-references external wind speed with internal mechanical sensors:
 - **Wind Gust (Normal Event):** High vibration + high wind speed + **normal gearbox temperature & oil pressure** $\rightarrow$ No false alarm raised.
 - **Mechanical Fault (Real Anomaly):** High vibration + normal wind speed + **elevated gearbox temperature or low oil pressure** $\rightarrow$ Flagged as an alert.
- **Status Badges:** Assets are automatically assigned a status: **NORMAL** (Green), **WARNING** (Yellow), or **CRITICAL** (Red).

3. AI Diagnostics & Revenue Loss Calculation

- **Revenue Loss Math:** When an asset is flagged, Python formulas calculate:
Daily Energy Loss (kWh) = Asset Capacity (kW) $\times$ Est. Downtime Hours $\times$ Degradation Factor%
- **Daily Revenue Risk (₹)** = Energy Loss (kWh) $\times$ Local Tariff Rate

LLM Diagnostic Summaries: The flagged sensor reading is passed to the **Google Gemini API**, which generates:

1. A 2-sentence explanation of *why* the asset is at risk (e.g., dust hotspotting or bearing wear).
2. A step-by-step repair checklist for field technicians.

4. Real-Time Web Interface & Demo Controls

- **Streamlit Framework:** Built in pure Python using **streamlit** and **plotly** for clean tables and visual graphs.
 - **Live Fault Slider:** The sidebar includes interactive sliders allowing your team to artificially inject a fault (e.g., cranking soiling to 45% or vibration to 80 Hz) live during judging.
 - **Instant UI Reaction:** Moving a slider instantly turns the asset badge red, updates the live telemetry graph, calculates the daily loss in ₹, and generates the AI diagnostic guide.
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4. Key Features:

S.No.	Feature Name	Description & Purpose
1	Simulated IoT Telemetry Tracking:	Ingests real-time operational readings across solar panels and wind turbines, including surface temperature, output current, voltage, power output, panel soiling percentage, turbine vibration, and gearbox temperature.
2	Rule-Based Anomaly Flagging & Health Scoring	Evaluates incoming sensor parameters against operating thresholds to automatically assign assets a real-time health index: NORMAL (Green) , WARNING (Amber) or CRITICAL (Red) .
3	LLM-Powered Root Cause & Diagnostic Summaries	Translates anomalous telemetry spikes into clear, natural-language diagnostic reports (e.g., explaining that high soiling and surface temperature indicate panel hotspotting) and provides step-by-step inspection checklists for field technicians.
4	Financial & Energy Loss Estimator	Calculates the estimated daily energy generation loss (in kWh) and monetary financial risk (in ₹).
5	Impact-Based Task Prioritization	Sorts and ranks flagged maintenance alerts by daily financial risk, ensuring field technicians prioritize high-loss, revenue-draining assets first.

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5. Technology & Implementation Plan

Specify the tech stack and phase-wise implementation roadmap:

Layer / Component	Technologies Used	Role in Project
Programming Language	Python	Core backend, scripting, and data processing
Frontend / Dashboard	Streamlit	Interactive web application & real-time telemetry charts
AI / Intelligence Engine	LLM API	Natural-language diagnostic generation & repair guides
Data Handling	Pandas, NumPy	Synthetic sensor telemetry creation & data manipulation

